GenDist: Generative-Driven Knowledge Distillation for Small Object Detection in Remote Sensing Images

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Abstract*—Optimal efficiency is always the primary pursuit of the remote sensing community. However, the blurred appearance in small objects causes increasingly deeper and wider detectors, posing a significant challenge for deployment. knowledge distillation (KD), which can shorten the time from image acquisition to interpretation, has become a hotspot. In this article, contrary to traditional feature imitation methods, we propose generative-driven KD for small object detection (GenDist), which reconstructs teacher-like representations from the partially masked student features to achieve knowledge transfer. Masking during training strengthens the student's ability to generalize in complex scenes. Specifically, GenDist contains three parts: online self-correct mask (O-SCM), dynamic activation global context (DAGC), and sampling-aligned generation (SAG). First, O-SCM is utilized to identify informative masking regions that guide the student to reconstruct the teacher-like features. By integrating an online self-correction mechanism, it adaptively directs generative supervision toward regions that are ignored or misinterpreted by students. Then, DAGC refines teacher features via global context modeling and critical feature selection, facilitating more accurate object detail capture by the student model. Finally, SAG improves the quality of feature reconstruction by introducing the learnable offset sampling, enabling students to learn irregular areas such as the small instances' edges and texture transitions. Our proposed method finds the sweet spot between precision and speed in an elegant way. GenDist yields stable performance gains for student models across the DIOR, AI-TOD, SODA-D, and SSDD datasets. For the lightweight detectors, it cuts the model's parameter count by roughly 38.2%, while keeping accuracy loss under 3.9%.

***Index Terms*—Knowledge distillation (KD), lightweight detectors, remote sensing images, small object detection.**

I. Introduction

SMALL objects are ubiquitous in remote sensing scenarios, including emergency response, ecological monitoring, and military surveillance [1], [2], [3], [4]. Regrettably, to address information loss and noisy representations for small objects, existing detectors frequently come at the cost of complex structure and heavy computational budgets, which restrict their deployment on resource-limited intelligent terminals, e.g., handheld smart devices and satellite on-orbit processing systems [5].

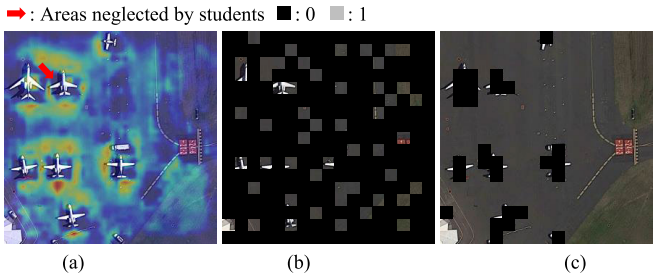


Fig. 1. Comparison of different masking strategies for feature reconstruction, where ZERO and ONE represent the features that are masked and preserved during generation, respectively. (a) Student networks may fail to capture critical foreground objects. (b) Random masking forces the student to reconstruct substantial irrelevant background information (i.e., black pixels-covered area). (c) Our O-SCM is designed to reconstruct features surrounding the foreground objects that the student network tends to miss. Best viewed in color.

Although pruning [6], [7], [8], quantization [9], [10], [11], and specific network structures [12], [13], [14] can produce lightweight detection models, there is always a gap in precision between compressed models and large models, due to the limited feature extraction capabilities. In contrast to these approaches, knowledge distillation (KD) trains compact student models under the guidance of powerful teacher models, significantly reducing parameters while maintaining teacher-like performance [15].

Recently, numerous KD-based works in remote sensing have emerged to achieve lightweight detection [17], [18], [19], [20], [21]. Such as, ScaleKD [22] and NKDFD [23] apply scale feature distillation to decouple the foreground from the background, thereby facilitating nonuniform feature extraction and knowledge transfer. SCRKD [24] leverages high-resolution frameworks as teachers to direct the distillation of lower resolution students. DKD [25] leverages a customized perceptual loss to process size-variant instances in remote sensing for adaptive transfer. Nevertheless, such approaches rely on passive numerical imitation, lacking targeted learning for small-sized objects. By comparison, the generative KD introduces a reconstruction paradigm [26], [27]. It lever-

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TABLE I
COMPARISON OF THE EFFECTS OF DIFFERENT MASKING STRATEGIES ON DISTILLATION IN DIOR. THE STUDENT MODEL IS GFL RESNET-50

Teacher	Masking strategy		mAP
	Random Mask	Online Self-Correct Mask (ours)	
GFL [16]	-	-	71.0
ResNet-101	✓	-	75.1
(75.3 mAP)	-	✓	75.3 (**+4.3**)

ages random masking to compel the student framework to restore the teacher's full features, hence implicitly modeling feature-level contextual dependencies. Unfortunately, directly applying such random masking strategies to remote sensing is suboptimal. As illustrated in Fig. 1(a), the student framework in remote sensing scenarios struggles to focus on informative foreground regions due to the weak feature extraction. What's problematic, as presented in Fig. 1(b), without semantic guidance, random masking causes the student networks to reconstruct uninformative background areas rather than focusing on valuable foreground content.

Despite its application in vision tasks, generative KD remains largely underexplored in object detection. This work aims to explore whether a novel generative KD method can effectively detect small objects in remote sensing scenarios while reducing computational cost. To achieve this, two key challenges must be addressed as follows.

1) How to identify valuable features for restoration?

The key to identifying features worth restoring lies in mask strategies, which guide students to recover features in specified locations (i.e., zeroed-out areas). Yet, as shown in Fig. 1(b), the position-agnostic binary random hard masking, as its lack of semantic awareness, produces highly stochastic distributions that have a tendency to ignore valuable foreground and instead reconstruct excessive redundant background [26]. In addition, the sharp boundaries of the binary hard masks disrupt contextual continuity, complicating the reconstruction of fine details in small-sized objects. Accordingly, designing a masking strategy that prioritizes informative areas and preserves contextual smoothness is of critical importance for knowledge transfer.

Given these issues, a natural idea is to reconstruct teacher-like features in areas where the student exhibits low attention. Thus, as presented in Fig. 1(c), we propose the online self-correct mask (O-SCM), which utilizes the student's online information (i.e., activation maps) as a reference to dynamically identify low-activation regions with weak semantic representations, thereby directing the learning focus toward neglected small objects. As shown in Table I, O-SCM consistently outperforms the random masking-based method [26] and equals the accuracy of the teacher framework using the lightweight backbone. Furthermore, we introduce a Gaussian-driven mask, which simulates real-world degradation using a center-to-edge decaying weight distribution. Such a soft weighting preserves activation in perturbed small object regions, providing residual information that guides feature reconstruction. O-SCM focuses the generative supervision on these under-attended areas, promoting more targeted and efficient knowledge transfer.

2) How to strengthen the reliability of restored features?

Existing generative KD [26] always treats teacher features as the ultimate reconstruction target. However, this assumption is not always held in complex scenes. Remote sensing images typically contain multiscale objects, complex textured backgrounds, and severe interclass ambiguity [5]. In such scenarios, teacher features often include substantial low-value redundancy or ambiguous semantic representations, such as imprecise responses to small instance boundaries or neglect of fine-grained textures [28], [29], [30]. Hence, the features reconstructed by student models during training may be unreliable.

To this end, we propose dynamic activation global context (DAGC) to capture long-range semantic dependencies, thereby rectifying unreliable supervision signals (i.e., teacher features) arising from the lack of global information for small instances. By recalibrating channel features via interpixel dependencies and injecting the global context with a selection mechanism, it transforms the ambiguous teacher features into high-quality supervision signals. Then, to improve feature reconstruction's spatial precision and robustness, we introduce the sampling-aligned generation (SAG) in the student model, implemented with the offset sampling. It adaptively explores object-specific receptive fields, promoting the reconstruction quality for complex and irregular instances.

Notably, these modules are used only for training and can be discarded at inference, avoiding any additional computational or storage cost. In summary, the contributions of this work can be summarized as follows.

1) A novel generative-driven distillation pipeline for small object detection is proposed, aiming to boost accuracy in complex scenarios while offering a robust and elegant baseline for low-latency remote sensing image processing terminals.
2) To resolve the limitations of random masking, O-SCM is introduced to adaptively focus generative supervision on regions where the student underperforms. Combined with a Gaussian-driven mask that preserves small object signals, the proposed masking strategy facilitates feature recovery in remote sensing scenes.
3) We propose DAGC to refine teacher features as higher quality learning targets by modeling global context and nonlinear activations. Besides, we introduce SAG in students to adaptively align with teacher feature boundaries via offset sampling, reinforcing the reliability of feature reconstruction.
4) Extensive experiments are carried out on four challenging small object datasets, i.e., DIOR, AI-TOD, SODA-D, and SSDD. Our GenDist achieves consistent gains for various student detectors, and outperforms existing state-of-the-art distillation methods.

This article is organized as follows. Section II reviews the related works, Section III introduces the proposed methodology, Section IV discusses the experimental results, and Section V concludes the article.

II. RELATED WORK

A. Small Object Detection

As one of the hottest topics, small object detection serves a crucial role in computer vision. Major challenges lie in limited resolution and poor semantic representation.

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To preserve small object local details and abstract semantics, feature pyramid network (FPN) [31] establishes a top-down pyramid architecture for feature fusion, facilitating the capture of discriminative information. Numerous works on FPN-based variants are then proposed. EFPN [32] utilizes subpixel convolution and upsampling operations to generate high-resolution features enriched with small object details. LFPN [33] solves the feature scale discrepancy problem in object detection with a multiscale latent space fusion and cross-modal alignment. RaFPN [34] complements the diluted features of small objects with cross-scale aggregation. However, these methods are optimized for natural images and rarely consider remote sensing.

In response to the challenges of small object detection in remote sensing scenes, ABNet [35] introduces channel attention to improve the backbone's feature representation and alleviate foreground occlusion caused by complex backgrounds. CSFF [36] captures interchannel dependencies to produce a robust and discriminative multilevel representation. SCRDet++ [37] denoises feature maps at the instance level to strengthen small object detection performance. ASSD [38] introduces a context-aware alignment module to correct mismatches between kernels and objects, enabling the extraction of robust, well-aligned features. To cope with more challenging scenarios, KLDet [39] replaces IoU with a distribution similarity metric to formulate a more adaptive label assignment method better suited for tiny-sized objects (i.e., less than 16×16 pixels).

Nevertheless, the heavy requirements on memory and computational resources limit the deployment of these detectors in resource-limited platforms. Hence, in this article, we are eager to develop detectors that strike a tradeoff between complexity and resource efficiency.

B. Detection-Oriented KD

KD [15] compresses models by transferring knowledge from a powerful teacher to a simpler student, achieving similar precision with fewer parameters and lower computation.

Research on this domain can typically be categorized into two directions: logical mimicry and feature mimicry. Logical mimicry focuses on adapting the decision logic of the network of teachers and students. It consists of mimicking categorical logit and bounding box regression. For the first time, Hinton et al. [15] transferred the dark knowledge from a powerful large teacher network to a small student network for classification. Zheng et al. [40] introduced a distillation loss for the regression task, and uses logical imitation by treating the teacher's discretized distribution as supervision to complete the training. Yin et al. [41] achieved adaptive regulation of interlayer differences via learnable slack variables. Albeit efficient, logical imitation relies heavily on the teacher's architecture, limiting its adaptability to heterogeneous distillation. Feature mimicry aims to distill intermediate details, e.g., rich multiscale features from the backbone or FPN. Zhang and Ma [30] and Yang et al. [29] introduced attention mechanisms in KD, so that student models adaptively focus on valuable regions in teacher networks. FGFI [42] and GID [43] both favor distilling features within the GT. FGFI encourages students to capture feature responses around object areas, while GID identifies regions with large teacher–student discrepancies as targets for distillation. However, these methods neglect to optimize for complex remote sensing scenarios, particularly in cases of limited object size or instances that are embedded in the background.

Several methods are proposed for KD in remote sensing. To select high-quality instances for knowledge transfer, Zhang et al. [25] integrated global distillation with a state-aware loss to dynamically filter informative samples. In a similar way, Yang et al. [17] captured valuable instances by modeling cost probability distributions and using multivariate knowledge mining. Yang et al. [44] designed a dynamic interactive learning model to boost the low-quality instances learning from high-quality ones. To fix the hierarchical misalignment, Zhang et al. [18] introduced a calibration and decoupling technique for adaptive cross-scale transfer. InsDist [19] disentangles instance-relevant foreground and instance-irrelevant background, promoting the learning of intraclass compactness and interclass separability. DPGD [20] facilitates the student's perception of feature maps distribution across different categories by capturing knowledge of global relations. ACFG [21] boosts the robustness of feature pixels under composite masks by an adaptive feature mapping. AFD [45] facilitates local distillation via introducing a multiinstance attention mechanism that distinguishes between background and foreground elements.

Existing feature mimicry-based KD requires the students to directly fit the teacher's intermediate features, which can result in rote imitation and a lack of proactive modeling capability. In addition, they ignore the risk of small instances being virtually wiped out, and the students are not allowed to freely leverage advanced teacher-like features. Hence, we introduce generative KD, compelling the student to infer holistic information from partial inputs and directing it to reconstruct high-value, robust features via composite masking.

III. Methodology

The overall framework is illustrated in Fig. 2. To be precise, the data flow in our framework is as follows: SAG reconstructs the student features masked via O-SCM, while DAGC refines teacher features to serve as the supervision target.

A. Preliminaries

During the training process of KD-oriented object detection, the input image is fed to a pair of object detectors: a learnable student and a frozen teacher. Existing paradigms always adopt the "backbone-FPN-head" pipeline. FPN is employed to model multiscale features of objects for subsequent classification and localization. Broadly, feature-based mimicry is defined as

$$\mathcal{L}_{\text{fea}} = \frac{1}{\hat{N}} \cdot \sum_{k=1}^{C} \sum_{i=1}^{H} \sum_{j=1}^{W} \hat{M} \cdot \left(\mathcal{F}_{k,i,j}^{t} - f\left(\mathcal{F}_{k,i,j}^{s} \right) \right)^2 \tag{1}$$

where H, W, and C indicate the height, width, and channel of the feature map, respectively. $\hat{N} = \sum_{k=1}^{C} \sum_{i=1}^{H} \sum_{j=1}^{W} \hat{M}$ is the normalization factor. The mask $\hat{M}$ is formulated variously

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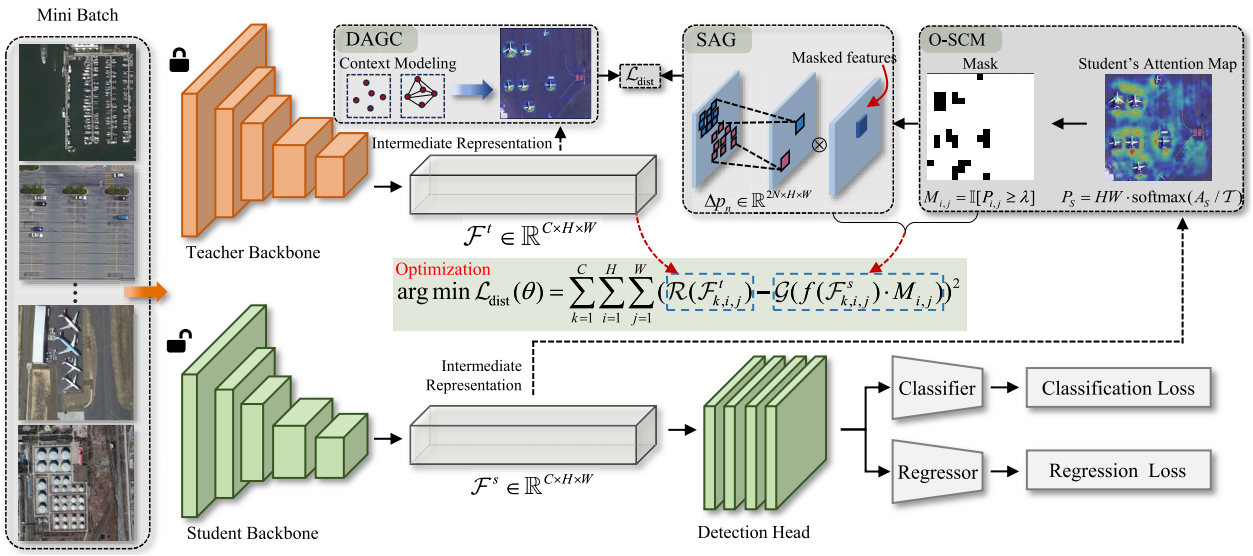


Fig. 2. Pipeline of the proposed GenDist. Here, $\mathcal{T}$ is a hyperparameter utilized to adjust the student feature map distribution, λ denotes the activation threshold of the mask, $\mathbb{I}$ is the indicator function, and Δp_n indicates the coordinates for offset sampling. The symbols "lock" and "unlock" demonstrate that the teacher's backbone is frozen and the student's backbone is learnable during training phases, respectively. The GenDist framework is trained jointly in an end-to-end fashion.

across methods and serves to emphasize the foreground while suppressing background noise. $\mathcal{F}^t$ and $\mathcal{F}^s$ represent the multiscale features from powerful teacher and lightweight student frameworks. $f(\cdot)$ is the adaptive layer used to resize $\mathcal{F}^s$ to the same resolution as $\mathcal{F}^t$.

Feature imitation-based KD is simple and efficient, but rigid alignment often causes the student to inherit the teacher's noise or redundant features. Despite their flexibility in transferring knowledge, recent generative KD methods suffer from increased reconstruction difficulty for small object features since random binary hard masks. To this end, we propose GenDist, advancing prior methods by generative feature selection and reliable reconstruction. Here, we will clarify our method in detail.

B. Online Self-Correct Mask

Generative KD enables the student model to perform knowledge transfer by reconstructing the teacher's features. Thus, it is crucial to find the parts worth generating. In this article, we argue that areas where students lack attention or have biased semantic understanding are the key to improving distillation effectiveness. To this end, we propose an O-SCM that dynamically identifies regions of high reconstruction value, providing precise guidance for feature generation.

As shown in Fig. 2, given a student feature $\mathcal{F} \in \mathbb{R}^{C\times H\times W}$ from FPN. We employ a spatial collaborative attention mechanism to capture the weights A_S, to refine the use of discriminative information from the student network

$$A_S = \frac{1}{C} \cdot \left\langle \|\mathcal{F}_1\|_2^2, \ldots, \|\mathcal{F}_Y\|_2^2 \right\rangle \tag{2}$$

where $A_S \in \mathbb{R}^{1\times H\times W}$ is spatial-wise attention maps. $\|\mathcal{F}_Y\|_2^2$ is the squared L_2 norm of the yth input feature vector. $\langle\cdot\rangle$ refers to the spatial reshape, where individual values computed for each pixel are collected to reconstruct an $H \times W$ weight map. $(1)/(C)$ calculates the mean to avoid extreme variations.

Spatial mask P_S, generated by activation functions, reflects regions where the student exhibits insufficient attention, such as small objects or areas with blurred textures

$$P_S = HW \cdot \text{softmax}\left(A_S/\mathcal{T}\right) \tag{3}$$

where $\mathcal{T}$ is a hyperparameter in softmax$(\cdot)$ [15] that modulates the distribution of the attention mask P_S: lower values sharpen the attention focus on confident locations, while higher values broaden the attention to encompass wider context. In O-SCM, attention scores reflecting student interest are binarized based on a threshold λ (0 if $< \lambda$, else 1). It can be expressed as

$$B_{i,j} = \begin{cases} 0, & \text{if } P_{i,j} < \lambda \\ 1, & \text{otherwise} \end{cases} \tag{4}$$

where λ set to 1, (i, j) denotes the coordinates on the feature map. Based on (4), insufficient attention parts are masked with 0, and their corresponding teacher's features function as generative supervision targets to guide student reconstruction. Then, to simulate realistic degradation scenarios, we propose a Gaussian-driven soft mask.

For the horizontal bounding box $H = \{x_0, y_0, w, h\}$ (central point coordinates, width and height), the value of 2-D Gaussian soft mask $E \in \mathbb{R}^{H\times W}$ at the spatial coordinate (i, j) can be formulated as

$$E_{i,j} = \begin{cases} e^{-\frac{(i-x_0)^2}{2(\sigma_x)^2} - \frac{(j-y_0)^2}{2(\sigma_y)^2}}, & \text{if } (i,j) \in H \\ 0, & \text{otherwise} \end{cases} \tag{5}$$

where $\sigma_x = w/2$, $\sigma_y = h/2$. A feature point that falls into the box H will retain its response. Gaussian-driven mask utilizes a center-to-edge decaying weight distribution, retaining a certain level of activation at occluded centers. Unlike the binary mask, which severs spatial continuity, soft masks preserve the fuzzy boundary between objects and context, allowing the generator

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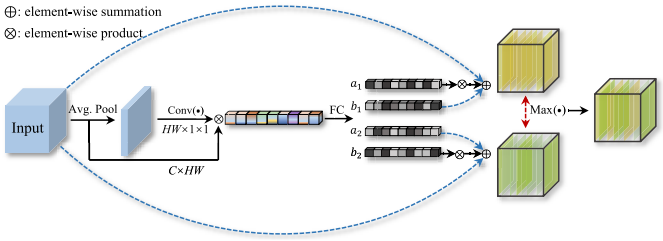


Fig. 3. Illustration of DAGC. By modeling global context and selecting essential channels, the teacher's intermediate features are effectively refined.

to infer object shape, location, and semantics from surrounding regions.

Naturally, we utilize this superior scheme to student models via combined masks

$$M = \alpha \cdot B + (1 - \alpha) \cdot E/Z \tag{6}$$

where α is a control parameter, balancing the influence of hard and soft masking. $Z = \sum_{i=1}^{H} \sum_{j=1}^{W} E_{i,j}$ is the normalization parameter. Equation (6) utilizes the dynamically updated activation maps to guide the students in reconstructing features at semantically weak regions, while using certain prompts to strengthen the specificity of the distillation procedure.

C. Dynamic Activation Global Context

Since the teacher features as the supervision signal, their quality directly determines the performance of distillation. Despite its superior expressive capability, the teacher networks inevitably struggle to capture small instances and their context details in a complex optical remote sensing condition, which hampers the student's ability to reconstruct critical parts effectively. Hence, we introduce the DAGC module to refine the teacher's feature map and provide students with more reliable learning guidance.

As presented in Fig. 3, we utilize DAGC to refine local and global features of teachers. Conceptually: 1) refine local details via an attention mechanism; 2) dynamically switch the feature channels ON and OFF to identify the most appropriate global representation.

Given a teacher feature $\mathcal{F} \in \mathbb{R}^{H \times W \times C}$ of a certain layer of FPN, the global embedding is obtained by

$$\mathcal{R}'(\mathcal{F}) = \sum_{j=1}^{N_P} \text{softmax}\left(W_k \mathcal{F}_j\right) \cdot \mathcal{F}_j \tag{7}$$

where $\mathcal{R}'(\mathcal{F}) \in \mathbb{R}^{C \times 1 \times 1}$, W_k is a convolutional layer, and N_P is the number of pixels in the feature. $\mathcal{R}'(\mathcal{F})$ fuses the contextual features from all channels to generate the refined vector with global dependencies. softmax$(\cdot)$ normalizes importance scores in the spatial dimension.

After that, $\mathcal{R}'(\mathcal{F})$ is fed into an activation layer consisting of two fully connected (fc) layers and activation functions to obtain the channel weights

$$[a_1, b_1, a_2, b_2]^T = \sigma\left(\text{fc}_2\left(\gamma\left(\text{fc}_1\left(\mathcal{R}'(\mathcal{F})\right)\right)\right)\right) \tag{8}$$

where σ and γ denote HSigmoid$(\cdot)$ and ReLU$(\cdot)$, respectively. $\text{fc}_1(\cdot)$ is used to compress the dimensions, while $\text{fc}_2(\cdot)$ is used to broaden the dimensions to $4C$. Equation (8) puts forward

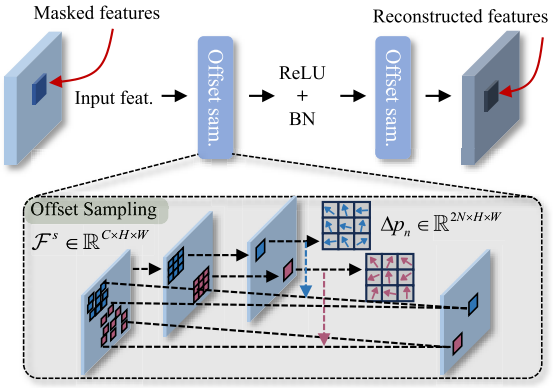


Fig. 4. Schematic of our SAG, where N is the number of sampling points in the convolution kernel. SAG acts on masked student features to recover teacher-like representations.

a nonlinear transformation that strengthens the expressiveness of the weight vector.

Ultimately, the DAGC module $\mathcal{R}(\cdot)$ adopts a set of learnable linear transformations to adaptively select the most appropriate features. It can be formulated as

$$\mathcal{R}(\mathcal{F}) = \mathcal{F} \cdot \max\left(a_1 \cdot \mathcal{R}'(\mathcal{F}) + b_1,\ a_2 \cdot \mathcal{R}'(\mathcal{F}) + b_2\right) \tag{9}$$

where $\max(\cdot, \cdot)$ as a dynamic switch over channel-wise feature slices. It automatically determines the appropriate set of linear transformations to refine teacher features (i.e., supervision signals). The input-dependent mechanism in our proposed DAGC facilitates the feature refinement flexibility and strengthens the fine-grained semantic expression in teachers.

D. Sampling-Aligned Generation

Generative blocks are introduced as a bridge for KD, guiding the student to actively reconstruct features under missing information. Specifically, we use flexible offset sampling to capture small object edges and textures, boosting feature reconstruction quality.

Key aspects of this section focus on the principle of SAG. To begin with, we review standard 2-D convolution, which consists of two phases: 1) sampling over the input features by the regular grid $\mathcal{K}$ and 2) summation of sampled values weighted by w

$$\mathcal{K} = \left\{\left(-\left\lfloor\frac{k}{2}\right\rfloor, -\left\lfloor\frac{k}{2}\right\rfloor\right), \left(-\left\lfloor\frac{k}{2}\right\rfloor, 0\right), \ldots, \left(\left\lfloor\frac{k}{2}\right\rfloor, \left\lfloor\frac{k}{2}\right\rfloor\right)\right\} \tag{10}$$

where $\lfloor\cdot\rfloor$ denotes the floor function. Its physical interpretation is to construct a square sampling grid centered at p_0. For each position p_0 on output features can be formulated as

$$\hat{\mathcal{W}}(x) = \sum_{p_n \in \mathcal{K}}^{N} w(p_n) \cdot x(p_0 + p_n) \tag{11}$$

where $N = k \times k$ denotes the size of the convolution kernel, x refers to inputs, p_n denotes the nth prespecified offset and w is the weight of kernel. Fixed square sampling grids $\hat{\mathcal{W}}$ cannot flexibly conform to the geometry of irregular small objects.

As illustrated in Fig. 4, beyond fixed sampling, we introduce a set of learnable offsets to the rigid grid $\mathcal{K}$ to achieve adaptive

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sampling within the different coordinates, and (11) can be reformulated as

$$\tilde{\mathcal{W}}(x) = \sum_{p_n \in \mathcal{K}}^{N} w(p_n) \cdot x(p_0 + p_n + \Delta p_n) \tag{12}$$

where each of the Δp_n is a tuple value (h, w), which is learned from the input features through a separate convolutional layer. Δp_n is implemented by a deformable convolution [46]. Due to the Δp_n denotes continuous variables, we apply differentiable bilinear interpolation $I(\cdot, \cdot)$ to align the features. The gradient *w.r.t.* the offset, represented as $(\partial \tilde{\mathcal{W}}(x))/(\partial \Delta p_n)$, is obtained by the chain rule on $I(\cdot, \cdot)$, thus, $\tilde{\mathcal{W}}$ aiding the learning of adaptive sampling in the network.

Equation (12) empowers the student network with a free-viewing capability, enabling a more interpretative absorption of teacher knowledge, especially in regions with structural complexity or scale variation. As depicted in Fig. 4, the pipeline of SAG is as follows:

$$\mathcal{G}(x) = \tilde{\mathcal{W}}_2\left(\text{ReLU}\left(\tilde{\mathcal{W}}_1(x)\right)\right) \tag{13}$$

where $\mathcal{G}(\cdot)$ represents the SAG module, which consists of two convolutional layers $\tilde{\mathcal{W}}_1$ and $\tilde{\mathcal{W}}_2$, along with the ReLU activation layer. The block plays a pivotal role in generative KD by learning to reconstruct the teacher features from the masked student inputs. It not only bridges feature recovery but also enhances the representational capacity of the student model, enabling more effective semantic transfer and structural alignment in complex visual tasks.

E. Loss Function

With the introduction of teacher supervision signals, the loss function $\mathcal{L}_{\text{dist}}$ of KD is defined as follows:

$$\mathcal{L}_{\text{dist}} = \sum_{k=1}^{C}\sum_{i=1}^{H}\sum_{j=1}^{W}\left(\mathcal{R}\left(\mathcal{F}^t_{k,i,j}\right) - \mathcal{G}\left(f\left(\mathcal{F}^s_{k,i,j}\right) \cdot M_{i,j}\right)\right)^2 \tag{14}$$

where O-SCM applies a mask M to the student feature $\mathcal{F}^s$ to erase informative regions. Then, SAG reconstructs the features by adaptive offset sampling $\mathcal{G}(\cdot)$. Concurrently, DAGC refines the teacher feature $\mathcal{F}^t$ to act as the supervision target $\mathcal{R}(\mathcal{F}^t)$.

The total loss $\mathcal{L}_{\text{total}}$ combines the student model's original loss $\mathcal{L}_{\text{det}}$ with the distillation loss $\mathcal{L}_{\text{dist}}$

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{det}} + \beta \cdot \mathcal{L}_{\text{dist}} \tag{15}$$

where β is a hyperparameter used to balance losses. $\mathcal{L}_{\text{det}}$ is the original loss of the student detector, which usually consists of a classification loss for category prediction and a regression loss for bounding box localization.

Compared with existing feature imitation-based distillation losses that depend on numerous balancing parameters, GenDist requires fewer hyperparameters, promoting stable training and better adaptability across architectures and visual tasks.

IV. Experiment

In this section, we begin by introducing four publicly available object detection datasets, along with the evaluation criteria and implementation specifics. Then, we conduct both quantitative and qualitative analyses of all experiments to demonstrate the effectiveness of our GenDist.

A. Datasets

1) DIOR: DIOR [47] is an ultralarge-scale remote sensing images dataset with 192 472 axis-aligned bounding box annotations across 23 463 images covering 20 object categories. To simplify, they are abbreviated as follows: airplane (AL), airport (AT), baseball field (BF), basketball court (BC), bridge (BG), chimney (CM), dam (DM), expressway service area (EA), expressway toll station (ES), golf course (GC), ground track field (GF), harbor (HB), overpass (OP), ship (SP), stadium (SD), storage tank (ST), tennis court (TC), train station (TS), vehicle (VH), and windmill (WM). The instances are categorized into three scales according to their absolute pixel areas: 1) small (S) with area $\leq 32^2$; 2) medium (M) with $32^2 <$ area $\leq 96^2$; and 3) large (L) with area $> 96^2$.

2) AI-TOD: AI-TOD [48] comprises eight prevalent object categories, 28 036 remote sensing images of 800×800 pixels with 700 621 object instances. The instances in an image are classified into four subscales according to their absolute pixel areas: 1) very tiny (vt) with $2^2 <$ area $\leq 8^2$; 2) tiny (t) with $8^2 <$ area $\leq 16^2$; 3) small (s) with $16^2 <$ area $\leq 32^2$; and 4) medium (m) with $32^2 <$ area $\leq 64^2$; and no large objects.

3) SODA-D: SODA-D [5] highlights the detection of those objects with extremely tiny resolution under complicated driving scenarios. Covering an extensive variety of environmental, perspectives and lighting conditions, it boasts a rich diversity of locations, climatic conditions, periods, shooting views, and scenes. Notably, it has 328 728 images of size 800×800 pixels. If we shed light on the smaller instances, the average absolute scale is just 20.31 pixels. SODA-D divides the small instances more finely: 1) extremely small (eS) with area $\leq 12^2$; 2) relatively small (rS) with $12^2 <$ area $\leq 20^2$; 3) generally small (gS) with $20^2 <$ area $\leq 32^2$; and 4) normal with area $> 32^2$.

4) SSDD: SSDD [49] is the first publicly available dataset for ship detection by synthetic aperture radar (SAR) imagery, widely utilized in deep learning-based studies. It spans resolutions from 1 to 15 m and covers different ship sizes, materials, and sea conditions, including both the nearshore and offshore conditions. SSDD contains 1160 images, with 2456 ships identified. For the experiment, 928 images are randomly selected for training, and 232 images are used for validation.

B. Evaluation Metrics

Following the evaluation protocols in MS COCO [50], we utilize the average precision (AP) to evaluate the performance of detectors. The relevant formulas are listed below

$$\text{AP} = \int_0^1 P(R)\ dR \tag{16}$$

where P and R refer to precision and recall. They are defined as follows:

$$\begin{aligned} P &= \frac{N_{TP}}{N_{TP} + N_{FP}} \\ R &= \frac{N_{TP}}{N_{TP} + N_{FN}} \end{aligned} \tag{17}$$

where N_{TP}, N_{FP}, and N_{FN} are the number of true positive, false positive, and false negative samples, respectively. As the

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TABLE II

QUANTITATIVE COMPARISON OF THE STATE-OF-THE-ART METHODS WITH OUR PROPOSED GENDIST ON THE DIOR DATASET. RESULTS WITH BOLD RED AND BLUE COLORS INDICATE THE BEST AND 2ND-BEST RESULTS OF EACH COLUMN, RESPECTIVELY. 20E: TRAINING 20 EPOCHS. †: REMOTE SENSING-SPECIFIC OBJECT DETECTORS

Method	Backbone	AL	AT	BF	BC	BG	CM	DM	EA	ES	GC	GF	HB	OP	SP	SD	ST	TC	TS	VH	WM	mAP
Generic detectors:																						
RFLA† [51]	ResNet-50	69.3	75.3	76.9	87.2	43.5	75.0	62.2	77.8	68.3	77.6	79.3	51.6	57.3	91.1	62.5	74.9	87.4	54.8	56.1	89.9	70.9
ASSD† [38]	VGG-16	85.6	82.4	75.8	89.5	40.7	77.6	64.7	67.1	61.7	80.8	78.6	62.0	58.0	84.9	76.7	65.3	87.9	62.4	44.5	76.3	71.1
FSoD-Net† [52]	ResNet-50	88.9	66.9	**86.8**	90.2	45.5	79.6	48.2	86.9	**75.5**	67.0	77.3	53.6	59.7	78.3	69.9	75.0	**91.4**	52.3	52.0	**90.6**	71.8
HawkNet† [53]	ResNet-50	65.7	84.2	76.1	87.4	45.3	79.0	64.5	82.8	72.4	50.2	82.5	**74.7**	59.6	89.7	66.0	70.8	87.2	61.4	52.8	88.2	72.0
ABNet†$_{20e}$ [35]	ResNet-50	66.8	84.0	74.9	87.7	**50.3**	78.2	67.8	85.9	74.2	79.7	81.2	55.4	61.6	75.1	74.0	66.7	87.0	62.2	53.6	89.1	72.8
AGMF-Net† [54]	ResNet-50	**90.9**	72.8	79.3	89.7	44.7	**81.4**	59.3	66.0	62.7	73.8	79.2	65.0	61.7	**91.7**	78.6	**75.8**	90.7	60.0	**58.0**	83.1	73.2
DetectoRS [55]	ResNet-50	60.1	**91.1**	75.5	87.8	49.6	80.4	**74.7**	86.3	72.9	84.6	84.0	60.8	**64.0**	72.9	74.2	59.9	86.9	**70.1**	44.9	86.6	73.4
KLDet†$_{20e}$ [39]	ResNet-50	67.5	81.1	75.5	88.2	47.4	78.6	65.0	82.9	72.7	79.9	80.8	57.1	61.2	**91.2**	74.6	75.2	87.4	56.9	56.0	**90.2**	73.5
DMOD† [56]	ResNet-50	**92.6**	85.5	84.0	89.7	**51.4**	80.2	68.7	70.2	64.3	83.5	81.2	**66.1**	**63.7**	78.2	69.8	**75.5**	90.2	62.6	55.3	83.0	74.8
KD-based detectors:																						
LD [40]	ResNet-50	63.1	73.3	71.7	84.8	31.8	76.4	54.3	72.8	59.7	76.5	80.2	48.7	49.9	82.0	66.9	58.5	83.7	51.1	42.3	81.5	65.5
IaDis† [57]	ResNet-50	82.1	83.1	**85.5**	**92.2**	35.3	**84.5**	64.0	**89.3**	**79.0**	82.2	73.5	64.6	61.1	89.9	67.1	71.6	**90.8**	43.8	**57.7**	83.2	74.0
AMD [27]	ResNet-50	68.4	86.7	76.1	89.5	47.8	79.8	68.8	87.5	69.4	84.8	83.6	61.1	63.1	89.2	76.7	71.1	89.7	64.3	51.4	88.0	74.8
FGD [29]	ResNet-50	67.8	87.0	77.1	89.9	47.6	78.8	69.0	86.4	70.6	**85.7**	**84.3**	60.9	62.9	89.3	**83.1**	70.1	89.1	63.6	51.7	88.4	**75.2**
MGD [26]	ResNet-50	69.2	**87.7**	76.3	89.9	47.4	79.7	69.5	87.9	71.2	**85.6**	83.2	61.6	62.7	89.5	78.4	70.2	89.7	63.6	51.7	88.4	**75.2**
GenDist (ours)	ResNet-50	69.1	86.9	77.3	**90.4**	47.7	79.2	**70.6**	**88.8**	73.1	85.3	**84.4**	62.2	62.7	89.8	**80.3**	70.7	89.7	**67.1**	51.9	89.0	**75.8**

paramount metric, the overall AP is obtained by averaging the AP over 10 IoU thresholds between 0.5 and 0.95 (with an interval of 0.05).

Several submetrics are used to evaluate model performance. AP_{50} and AP_{75} correspond to precision at IoU thresholds of 0.50 and 0.75. Moreover, to highlight the model's robustness to instances of different scales, we report AP for diverse area-based subsets, e.g., the AI-TOD includes AP_{vt}, AP_t, AP_s, and AP_m, which represent the detection performance on very tiny, tiny, small, and medium-sized objects, respectively.

To illustrate computational complexity, we present the relevant metrics, including parameters (#Param.), floating-point operations (FLOPs), and frames/s (FPS). Specifically, #Param. and FLOPs indicate the model size and computational cost, while FPS represents inference speed, with higher values indicating faster processing of a single image.

C. Implementation Details

All experiments are performed on the NVIDIA A100 GPU, with implementations based on the MMDetection toolbox [73] and PyTorch [74]. Following the practice [20], [21], for a fair comparison, all the KD-based methods in the main experiment utilize GFL [16] as the architecture for both the teacher and student models. The teacher's backbone is ResNet-101, trained for 24 epochs; the student's backbone is ResNet-50, trained for 12 epochs. For the student framework, if not stated specifically, all frameworks are trained using the stochastic gradient descent (SGD) optimizer with a momentum of 0.9, a mini-batch size of 8, and an initial learning rate of 0.01, which decreases by a factor of 10 at the $8th$ and $11th$ epoch. During the inference, the first 1000 bounding boxes are retained for the dense small object predictions. We report overall AP and fine-grained scale metrics to validate small object detection performance. Specifically, we evaluate AP_S, AP_M, and AP_L for DIOR and SSDD, and finer scales (e.g., AP_{vt}, AP_t, AP_{eS}, and AP_{rS}) for AI-TOD and SODA-D. We employ the following hyperparameter settings: $\lambda = 1.0$, $\alpha = 0.9$, $\beta = 1.0e^{-5}$, and $\mathcal{T} = 0.5$.

D. Comparison With State-of-The-Art Methods

1) Results on DIOR: The results of the comparative experiments are depicted in Table II. To validate the generalization of our GenDist, we both compare it with the remote sensing-specific detection frameworks and KD-based detectors. Results involved in the comparison come from the previous research [35] and the related publications [47]. As demonstrated in the results, our GenDist achieves the best overall performance with a 75.8 mAP. It also demonstrates competitive performance in representative categories of large/medium/small objects such as, "golf courses," "basketball court," and "windmills." The visualization results are demonstrated in Fig. 5. Compared to the heavy general object detector DetectoRS [55] and the most recently proposed KLDet [39], developed for remote sensing tiny object detection (i.e., less than 16×16 pixels), GenDist leads by approximately 2.3 mAP. In which, the heavy detector DetectoRS [55] benefits from the refinement stage, achieving competitive performance on large-scale objects. However, the complex architecture delays inference, which is not conducive to practical deployment. Furthermore, the recently lightweight detector KLDet [39] requires more training epochs. Compared with the remote sensing KD approach IaDis [57], our GenDist outperforms it with a clear margin of 1.8 mAP, effectively confirming its effectiveness in knowledge transfer. For the general KD-based frameworks, both MGD [26] and AMD [27] depend on a coarse masking strategy for feature generation, resulting in inferior results as they typically generate excessive redundant background information. In comparison, our GenDist achieves outstanding results across multiple categories, bringing comprehensive refinements to the student model.

2) Results on AI-TOD: In Table III, we compare GenDist with the state-of-the-art methods on the AI-TOD dataset [48]. The selected methods cover the typical single-stage and two-stage generic detectors, along with networks specifically developed for remote sensing scenarios. Despite being tailored for remote sensing and introducing localization metrics suited for tiny objects (i.e., smaller than 16×16 pixels),

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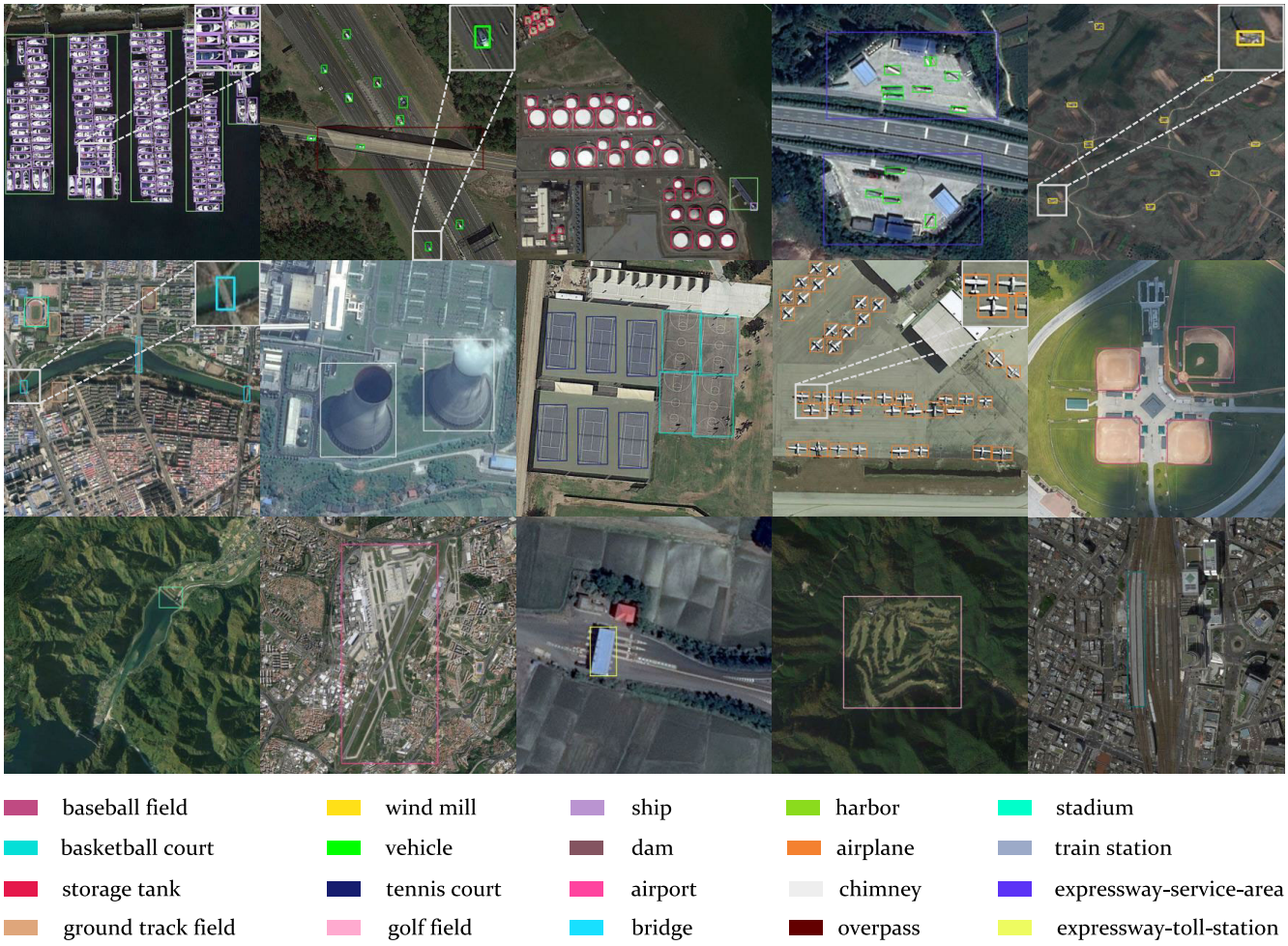


Fig. 5. Visualization results of GenDist in the large-scale cross-scale optical remote sensing dataset DIOR. Best viewed in color, and a zoomed-in view for details.

TABLE III

QUANTITATIVE COMPARISON OF THE STATE-OF-THE-ART METHODS WITH OUR GENDIST ON THE AI-TOD DATASET. ALL MODELS ARE TRAINED WITH RESNET-50. W/. P2: USING P2-P6 OF FPN

Method	w/. P2	AP	AP_{50}	AP_{75}	AP_{vt}	AP_t	AP_s	AP_m
Generic detectors:								
PKS RCNN† [58]	-	9.3	22.1	6.3	0.0	6.6	18.9	-
CEASC† [59]	-	14.4	33.2	10.2	3.2	15.2	18.5	-
DetectoRS [55]	-	14.8	32.8	11.4	0.0	10.8	**28.3**	**38.0**
DotD† [60]	-	16.1	39.2	10.6	8.3	17.6	18.1	22.1
RFLA† [51]	✓	16.3	39.1	11.3	7.3	18.5	19.8	21.8
KLDet† [39]	✓	19.6	46.8	13.3	**8.6**	**20.7**	22.8	25.9
Salience DETR [61]	-	19.7	48.4	12.7	7.4	19.7	25.9	-
FSANet† [62]	-	20.3	48.1	**14.0**	6.3	19.0	**26.8**	**36.7**
KD-based detectors:								
MGD [26]	-	11.9	27.8	8.9	0.1	8.6	24.5	35.1
AMD [27]	-	20.1	49.7	12.8	6.8	20.4	25.2	31.5
FGD [29]	-	**20.5**	**50.1**	13.0	**9.7**	20.5	25.6	31.7
GenDist (ours)	-	**21.3**	**51.8**	**13.6**	7.6	**21.5**	26.6	32.6

TABLE IV

QUANTITATIVE COMPARISON OF THE STATE-OF-THE-ART METHODS WITH OUR GENDIST ON THE SODA-D DATASET. ALL MODELS ARE TRAINED WITH RESNET-50 AS THE BACKBONE

Method	Epoch	AP	AP_{50}	AP_{75}	AP_{eS}	AP_{rS}	AP_{gS}	AP_N
Generic detectors:								
Deformable-DETR [63]	50	19.2	44.8	13.7	6.3	15.4	24.9	34.2
CenterNet [64]	70	21.5	48.8	15.6	5.1	16.2	29.6	42.4
CornerNet [65]	24	24.6	49.5	21.7	6.5	20.5	32.2	43.8
RFLA † [51]	12	29.7	**60.2**	25.2	13.2	26.9	35.4	44.6
CFINet [66]	12	**30.7**	**60.8**	26.7	**14.7**	**27.8**	36.4	44.6
KD-based detectors:								
MGD [26]	12	30.6	58.2	27.4	13.3	26.7	37.3	**46.0**
LD [40]	24	**30.7**	58.1	**27.6**	13.0	26.3	**37.8**	45.9
GenDist (ours)	12	**31.5**	59.3	**28.5**	**13.7**	**28.1**	**38.0**	**47.3**

RFLA [51] and KLDet [39] still underperform compared to GenDist in overall precision. Besides, they often rely on the higher resolution P2 layer in the FPN [31] to obtain competitive results, though this operation reduces inference speed. Among KD-based methods, GenDist achieves higher distillation performance when using the same student framework. For example, it achieves the best result of 21.5 AP_t on the tiny subcategory, outperforming FGD [29] with 1.0 AP_t. Compared with the generative distillation-based detector AMD [27], GenDist outperforms it by 0.8 AP_{vt} in the extremely tiny category.

3) Results on SODA-D: To validate the generalizability of GenDist in autonomous driving scenarios, we compare it with state-of-the-art methods in Table IV. From the results, in

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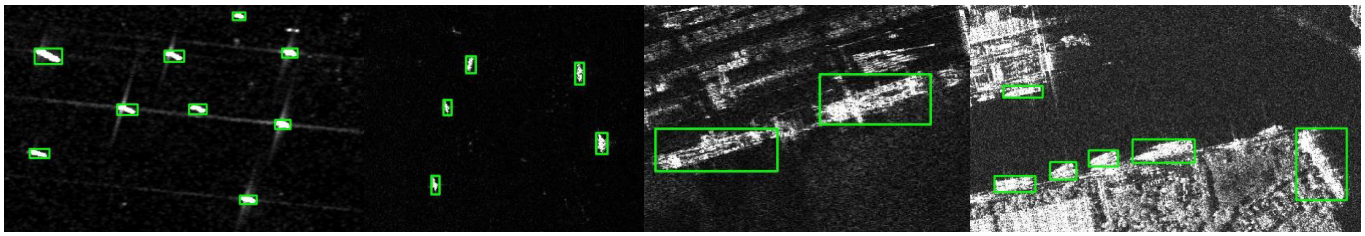


Fig. 6. Visualization results of our proposed GenDist on the SAR dataset SSDD.

TABLE V

QUANTITATIVE COMPARISON OF THE STATE-OF-THE-ART METHODS WITH OUR GENDIST IN THE SAR DATASET SSDD. ∗: MULTISCALE TRAINING. 2×: TRAINING 24 EPOCHS. INPUT IMAGE RESOLUTION: 1280 × 800 PIXELS

Method	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L	#Param. (M)↓	FPS↑
Generic detectors:								
DINO*$_{50e}$ [67]	54.9	78.7	64.1	53.7	60.9	44.9	47.54	17.8
VFNet [68]	62.3	92.0	71.8	64.2	56.4	40.7	32.48	25.1
RFLA†$_{2\times}$ [51]	62.5	**96.5**	73.5	63.8	58.3	25.1	41.12	28.0
AutoAssign [69]	65.1	97.1	78.3	66.4	61.8	21.0	35.97	27.2
FSAF [70]	65.1	94.9	78.9	67.2	59.6	27.4	36.01	27.0
DyHead [71]	65.3	94.8	80.1	65.7	64.9	**50.7**	39.07	18.9
GFL [16]	65.8	95.1	79.8	66.6	63.7	29.9	32.03	28.8
ATSS [72]	66.3	96.0	79.5	67.3	64.0	28.6	31.89	27.1
KD-based detectors:								
AMD [27]	44.0	71.9	49.0	51.4	16.6	35.4	32.08	26.4
LD*$_{2\times}$ [40]	44.3	71.8	51.4	48.8	30.1	6.9	32.03	27.0
MGD [26]	**68.7**	96.1	**83.3**	**68.6**	70.0	38.8	32.08	28.6
FGD [29]	**68.7**	**96.4**	83.0	68.1	**70.9**	50.2	32.03	28.6
GenDist (ours)	**69.1**	96.3	**83.6**	**68.8**	**70.6**	**54.6**	32.03	28.8

spite of longer training times, the AP of the general detectors CenterNet [64] and CornerNet [65] is only 21.5 AP and 24.6 AP, significantly inferior to our approach. We further compared our method with the recently popular transformer-based methods. Deformable-DETR [63] integrates deformable convolution [46] for sparse sampling and relational modeling capability of transformer [75]. It decreases the computational cost in the encoder and enhances detection accuracy for small objects by emphasizing important features in the feature map. Despite longer training time, Deformable-DETR [63] achieves only 19.2 AP, a result that is considerably lower than GenDist. This performance gap implies that the sparse query paradigm may not sufficiently capture tiny objects. In contrast, GenDist achieves an optimal performance of 31.5 AP with fewer training epochs. The reduced cost makes it more favorable for practical deployment. Importantly, our approach achieves the best AP_{75}, indicating high precision under strict constraints. Furthermore, GenDist's gains on small instances do not come at the cost of accuracy for normal-sized categories.

4) Results on SSDD: In Table V, we compare our GenDist with state-of-the-art approaches on the SAR dataset SSDD [49]. In terms of detection accuracy, our method outperforms classic approaches with the highest accuracy of 69.1 AP. In the challenging small object subset, our proposed method reaches 68.8 AP_S, demonstrating its ability to preserve fine-grained details. In contrast, LD [40], which reformulates regression as classification, only achieves 44.3 AP, likely since inaccurate teacher localization leading to failed distillation. In terms of model parameters and inference speed, our proposed GenDist requires only 32.03 M, making it smaller than generic detectors, e.g., DINO (47.54 M) [67] and AutoAssign (35.97 M) [69]. Vitally, GenDist achieves a high speed of 28.8 FPS, matching the fastest competitors and surpassing the accurate but slower DyHead (18.9 FPS) [71]. This demonstrates that our GenDist achieves competitive results while strictly maintaining its lightweight and high-speed nature, which is critical for the real-time deployment phase. The visualization results are presented in Fig. 6.

E. Ablation Study

To verify the effectiveness of each component, we perform a series of comparative experiments on the `test set` of DIOR [47] and SSDD [49]. Table VI illustrates the performance and cost of our GenDist over different conditions.

1) Effect of O-SCM: To verify the effect of O-SCM, we compare the performance improvements with and without this approach in Table VI. The results demonstrate that the O-SCM increases the detection accuracy of the student model by 4.3 mAP and 2.9 AP on the DIOR and SSDD datasets, respectively, with no additional parameters introduced. To further investigate the contribution of the binary mask B and Gaussian soft mask E within O-SCM on model performance, we modulate α within the range $[0, 1]$ (step size: 0.1) to yield different combinations. As shown in Fig. 7(a), relying solely on the soft Gaussian mask (i.e., $\alpha = 0$) fails to compel the student to perform complete feature reconstruction, resulting in suboptimal performance. In contrast, once the hard mask B becomes dominant, performance improves gradually, peaking at $\alpha = 0.9$. However, completely eliminating the soft mask (i.e., $\alpha = 1$) leads to a decline in AP and AP_S. Hence, the synergy between hard and soft masks yields reconstruction results superior to using either strategy in isolation.

2) Effect of DAGC: Within GenDist, DAGC is designed to improve the feature representation of teacher networks, offering superior supervision signals to the student model. As previously noted, the complex imaging conditions in remote sensing scenes can cause the original teacher features to be disrupted by ground clutter, while DAGC mitigates this challenge effectively through context modeling. As illustrated in Table VI, our DAGC achieves a 4.4 mAP gain (75.4 versus 71.0) and a 2.8 AP boost (68.6 versus 65.8) compared to the baseline, with negligible computational cost, validating the effectiveness of this component.

3) Effect of SAG: Given the challenge of accurately reconstructing the small object features, we introduce SAG with the offset sampling mechanism to instruct the student network

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TABLE VI

ABLATION STUDY ON THE DIOR AND SSDD DATASET. THE NUMBERS IN () ON THE LEFT MEAN IMPROVEMENTS OVER THE BASELINE, WHEREAS THE RIGHTMOST VALUE IS THE REDUCTION IN INFERENCE TIME OF GENDIST RELATIVE TO THE TEACHER

Method	O-SCM	DAGC	SAG	mAP on DIOR	AP on SSDD	#Param. (M) ↓	FLOPs (G) ↓	Per-image inference time (ms) ↓
Teacher	-	-	-	75.3	68.8	51.07	281.52	35.71
Student (baseline)	-	-	-	71.0	65.8	32.66	218.98	27.86
Ours	✓	-	-	75.3 **(+4.3)**	68.7 **(+2.9)**	32.66	218.98	27.86
	-	✓	-	75.4 **(+4.4)**	68.6 **(+2.8)**	32.74	219.28	27.86
	-	-	✓	75.4 **(+4.4)**	68.5 **(+2.7)**	33.27	221.45	27.86
	✓	✓	✓	75.8 **(+4.8)**	69.1 **(+3.3)**	33.35	221.75	27.86 **(-21.98%)**

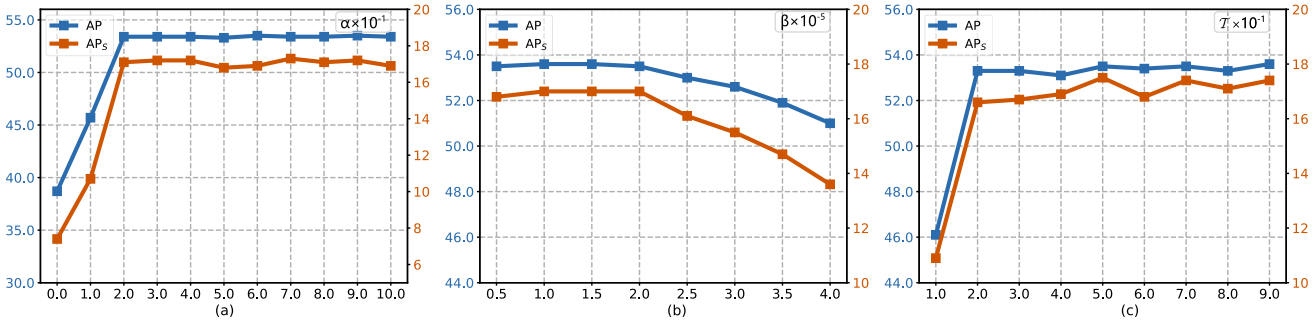


Fig. 7. Hyperparameter sensitivity study of (a) α, (b) β, and (c) $\mathcal{T}$ on the DIOR dataset. We report both the AP (left axis) and the small object subset precision AP$_S$ (right axis). Best viewed in color.

TABLE VII

COMPARISON BETWEEN GENDIST AND KD METHODS ON DIOR. THE BACKBONE INCLUDES BOTH CNN- AND TRANSFORMER-BASED ARCHITECTURES

Method	Faster RCNN [77]						RetinaNet [76]						FCOS [78]						Faster RCNN (Swin) [79]					
	AP	AP$_{50}$	AP$_{75}$	AP$_S$	AP$_M$	AP$_L$	AP	AP$_{50}$	AP$_{75}$	AP$_S$	AP$_M$	AP$_L$	AP	AP$_{50}$	AP$_{75}$	AP$_S$	AP$_M$	AP$_L$	AP	AP$_{50}$	AP$_{75}$	AP$_S$	AP$_M$	AP$_L$
Teacher	50.0	75.3	55.6	17.5	42.4	67.3	49.1	72.0	53.0	13.8	41.6	67.0	49.8	76.3	53.6	17.7	41.9	67.6	53.9	78.4	60.2	18.8	45.9	71.8
Student	41.3	69.4	43.0	13.2	34.3	57.1	37.1	60.2	39.4	8.4	29.7	52.7	39.4	67.0	40.3	10.4	32.3	56.1	49.1	75.3	54.1	15.0	38.3	66.5
AT [80]	43.1	70.6	46.1	14.1	34.9	59.4	38.3	61.6	41.7	8.6	30.1	54.8	40.8	65.7	43.3	10.4	32.8	57.4	49.8	77.0	54.5	16.9	41.8	66.1
KD [15]	43.4	70.7	46.3	13.6	35.2	59.7	38.7	61.7	42.0	8.7	29.2	54.8	41.2	65.8	43.4	10.3	33.1	58.0	50.3	75.9	55.6	14.0	39.8	68.8
AFD [45]	43.6	71.4	46.5	12.8	34.4	60.3	39.7	63.0	42.9	10.3	30.5	55.9	41.9	66.7	44.2	11.0	33.4	58.6	50.5	75.9	55.7	14.5	39.7	69.1
FGFI [42]	43.8	71.2	47.1	14.4	35.8	59.9	41.4	64.2	45.1	9.6	31.5	58.8	-	-	-	-	-	-	51.3	77.8	56.5	17.6	43.4	68.3
TWKD [81]	43.9	71.2	47.5	12.2	33.9	61.2	39.3	63.0	42.8	10.1	31.3	55.0	42.1	66.8	44.5	11.1	33.3	58.9	50.7	76.2	56.0	14.7	40.1	69.4
FKD [82]	44.2	71.7	47.7	14.1	35.9	60.4	41.7	65.9	45.5	10.6	33.6	58.1	42.2	67.4	44.6	10.3	33.3	59.8	51.0	76.8	57.2	15.1	40.3	70.0
DeFeat [83]	44.4	71.6	47.9	14.2	36.0	60.7	41.5	64.6	45.0	10.2	32.9	58.1	42.9	68.1	45.8	11.0	33.5	60.5	51.8	78.1	57.0	17.4	43.5	69.7
ARSD [28]	44.5	71.9	47.8	12.9	34.7	61.9	42.1	66.4	45.8	10.3	34.0	58.3	42.8	68.1	45.5	10.8	33.5	60.4	51.1	76.2	56.7	15.0	40.0	69.6
FRS [84]	44.9	71.3	48.9	13.9	34.9	62.5	42.4	66.0	45.9	10.5	33.5	58.7	42.4	67.6	45.1	10.8	33.2	59.9	51.6	76.6	57.2	15.9	41.9	69.5
InsDist [19]	45.1	72.4	48.5	12.8	34.9	62.7	43.0	66.7	46.4	10.9	34.0	59.5	43.0	68.2	46.1	11.4	33.7	60.1	51.9	78.0	57.3	16.9	44.0	70.6
DPGD [20]	46.9	73.7	51.0	14.4	36.6	64.7	45.1	69.5	48.7	12.3	36.5	62.0	44.8	70.3	47.5	12.4	35.6	62.1	53.9	78.6	60.0	18.0	45.1	72.2
GenDist (ours)	**48.1**	**73.0**	**52.8**	**14.6**	**37.8**	**65.7**	**47.0**	**71.6**	**51.0**	**12.7**	**37.5**	**64.9**	**47.5**	**73.3**	**50.9**	**13.6**	**37.0**	**65.8**	**53.9**	**80.1**	**60.3**	**19.5**	**44.7**	**70.3**

in achieving high-quality feature recovery. Our SAG achieves a 4.4 mAP improvement (75.4 versus 71.0) on DIOR [47] compared to the baseline, with a minimal increase of around 0.61-M parameters and 2.47-G FLOPs. It is important to note that these additional computational costs do not affect the inference speed, since they are solely introduced during the training and are not part of the inference period.

4) Hyperparameters: We investigate the sensitivity of our GenDist to four hyperparameters: λ, α, β, and $\mathcal{T}$. Specifically, λ is the mask threshold, α is used to alter the composite mask, β balances the distillation loss, and $\mathcal{T}$ modulates the attention mask distribution. As illustrated in Fig. 7, we present both the AP and AP$_S$. The results demonstrate that the composite mask formed with $\alpha = 0.9$ is optimal. When $\beta = 1.0e^{-5}$, the model produces competitive overall and small object precision, while $\mathcal{T} = 0.5$ achieves the highest AP$_S$ of 17.5. As λ fluctuates within the range of 0.8 to 1.2 (at intervals of 0.1), AP shows slight variation between 53.4 and 53.6. For simplicity, we set λ to 1.0. Thus, hyperparameters are 1.0, 0.9, $1.0e^{-5}$, and 0.5.

In summary, GenDist adopts an offline distillation paradigm where auxiliary modules (O-SCM, DAGC, and SAG) are discarded during inference. Thus, inference latency remains at the baseline (27.86 ms/img), ensuring no extra computational burden.

F. Exploratory Experiment

1) Comparison With the KD-Based Methods: We introduce GenDist in comparison with other distillation methods on the DIOR dataset. The results are depicted in Table VII. Following practice [20], the backbones of the teacher model and student model are ResNet-101 and ResNet-18. For Faster RCNN [76], the transformer architectures Swin-Base and Swin-Tiny [79] serve as the backbone networks for the teacher framework and student model, respectively. Moreover, the FGFI [42] is only

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TABLE VIII

FEW-SHOT OBJECT DETECTION IN DIOR. "#IMAGES": THE NUMBER OF TRAINING IMAGES. "PROP.": THE TRAINING IMAGES SAMPLING PROPORTION. THE STUDENT BACKBONE IS RESNET-50. PER-EPOCH TRAINING TIME IS RECORDED ON GEFORCE RTX 3090 (BATCH SIZE: 8)

#Images	Prop.	Method	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L	Train.(H)
11725	100%	Teacher	51.8	75.4	56.4	16.1	43.4	70.2	0.505
		Student	47.1	71.1	51.1	15.1	37.4	65.3	0.212
2391	25%	LD [40]	25.7	39.8	27.2	3.1	20.3	41.2	0.100
		DetectoRS [55]	34.6	56.4	36.1	6.2	27.2	52.3	0.164
		GenDist (ours)	**43.9**	**64.9**	**47.9**	**10.0**	**32.1**	**64.2**	0.117
5863	50%	DetectoRS [55]	42.5	63.5	46.3	7.9	32.3	62.3	0.326
		FGD [29]	47.9	70.1	51.9	11.0	37.1	67.8	0.191
		GenDist (ours)	**49.4**	**71.4**	**53.9**	**11.9**	**37.8**	**69.5**	0.233
8794	75%	LD [40]	43.4	63.6	47.0	9.1	33.3	63.0	0.294
		DetectoRS [55]	47.4	70.3	52.1	10.7	37.1	67.2	0.489
		GenDist (ours)	**52.2**	**75.2**	**56.9**	**15.1**	**41.2**	**71.5**	0.354
11725	100%	DetectoRS [55]	49.7	73.0	54.7	12.7	39.9	69.4	0.652
		LD [40]	45.8	66.7	49.5	10.4	35.4	65.1	0.400
		GenDist (ours)	**53.5**	**76.7**	**58.5**	**17.0**	**43.0**	**72.3**	0.472

compatible with anchor-based detectors and is not suitable for FCOS [78]. Based on the results, we conclude that: i) GenDist exhibits stronger cross-architecture generalization, consistently achieving substantial performance improvements on different detectors. On DIOR [47], it yields notable gains for the student models, with 6.8 AP, 9.9 AP, and 8.1 AP boosts. These gains are attributed to the utilize of self-correcting masks and refines in the generation module, which greatly augment the quality of generated feature maps. ii) On the popular transformer-based backbone Swin-T [79], our GenDist achieves 4.8 AP boost for Faster RCNN [76]. Particularly, GenDist outperforms the most recently proposed KD-based DPGD [20] 1.5 AP_S in the small object subset, proving its adaptability under strict conditions.

2) Training With Fewer Samples: We further verify GenDist performance on the few-shot detection task for small objects. The target of few-shot detection is to learn a network that can achieve generalization from only a few training samples [85]. Compared to regular object detection, this task presents extra difficulty due to the unreliable empirical risk of a few samples.

To conduct this experiment, we randomly select a specific proportion of samples from DIOR to create four corresponding subsets. The size of every subset and its proportion relative to DIOR are presented in the first two columns of Table VIII. For every subset, we train both our approach and the comparison methods from the beginning and present the detection results on the original full `test set`. As depicted in Table VIII, our suggested GenDist consistently outperforms DetectoRS [55] and LD [40] across all few-shot settings. Notably, when trained on merely 50% of the training samples, our GenDist achieves a 2.3 AP improvement over the student model trained on 100% of the DIOR dataset. Furthermore, under the identical training conditions, the training period required by GenDist is steadily shorter than that of DetectoRS [55], highlighting our GenDist's potential for real-world applications. These results underscore that GenDist is highly effective in few-shot settings and offers significant promise for situations where labeled data is severely limited.

TABLE IX

DISTILLATION EFFECTS OF OUR GENDIST ON THE DIOR DATASET FOR DIFFERENT LIGHTWEIGHT BACKBONES. "T": TEACHER NETWORK. "S": STUDENT NETWORK

Lightweight Model	Epoch	MS_{train}	AP	AP_S	#Param.(M)	FLOPs(G)	FPS
GFL-ResNet50T [16]	24	✓	51.1	16.2	32.08	215.44	37.3
GFL-ResNet34S [16]	12	-	44.8	10.4	29.16	193.62	49.4
GenDist (ours)	12	-	**50.7**	**13.7**	**29.85**	**196.39**	**49.4**
GFL-ResNet34T [16]	24	✓	50.0	14.6	29.16	193.62	49.4
GFL-ResNet18S [16]	12	-	40.9	8.0	19.13	155.79	60.8
GenDist (ours)	12	-	**49.1**	**12.6**	**19.82**	**158.56**	**60.8**

TABLE X

COMPARISON OF DIFFERENT MASKING MODES, EACH APPLIED TO THE MULTISCALE FEATURE LAYER OF FPN. "RATIO": RATIO OF COVERED FEATURES. "ADA.": MASK WITHOUT HAND-CRAFTED SETTINGS

Mask Strategy	Ratio	Loss	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L
No Mask (baseline)	-	L_2	47.1	71.1	51.1	15.1	37.4	65.3
Random Mask	60%	L_2	50.8	74.9	54.8	14.0	40.7	70.1
Grid Mask	60%	L_2	51.7	75.1	56.1	15.1	41.5	70.7
Center Mask	Ada.	L_2	51.6	74.9	56.0	15.0	41.3	70.5
Importance-based Mask	Ada.	L_1	51.0	74.5	55.6	15.0	41.3	69.8
Importance-based Mask	Ada.	L_2	51.4	75.0	55.9	15.3	41.1	70.1
O-SCM (ours)	Ada.	L_2	**52.3**	**75.8**	**56.8**	**15.6**	**42.1**	**71.4**

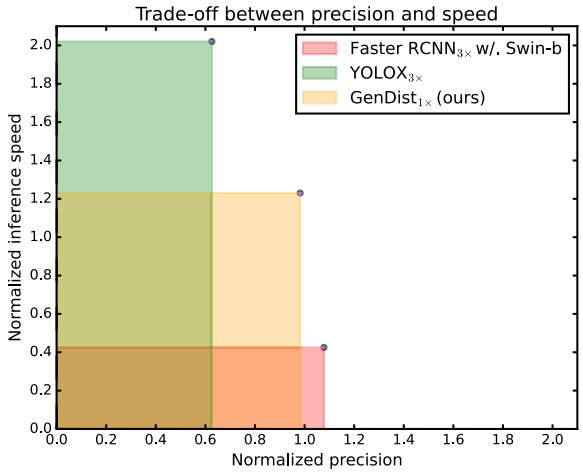


Fig. 8. Comparison of different frameworks for tradeoffs between inference speed and precision. 3× means training for 36 epochs. Best viewed in color.

3) Effect on Lightweight Backbone: In remote sensing image processing, real-time inference is critical, and the networks deployed on lightweight backbones must maintain competitive results. To validate the performance of our proposed approach with lightweight backbones, we also compare it with different networks in Table IX. Our proposed GenDist yields consistent performance gains across all student models without extra data augmentation, yielding AP gains of 5.9 and 8.2 in ResNet-34 and ResNet-18, respectively. Notably, our GenDist attains the largest relative gain on the compact ResNet-18 student model, achieving a **38.2%** parameter cut with a quantization precision loss within 3.9% compared to GFL-ResNet50T [16], confirming its effectiveness in transferring knowledge to lightweight backbones.

4) Exploration of Different Mask Designs: We implement a series of exploratory experiments to investigate how different masking strategies affect generative KD. As reported in

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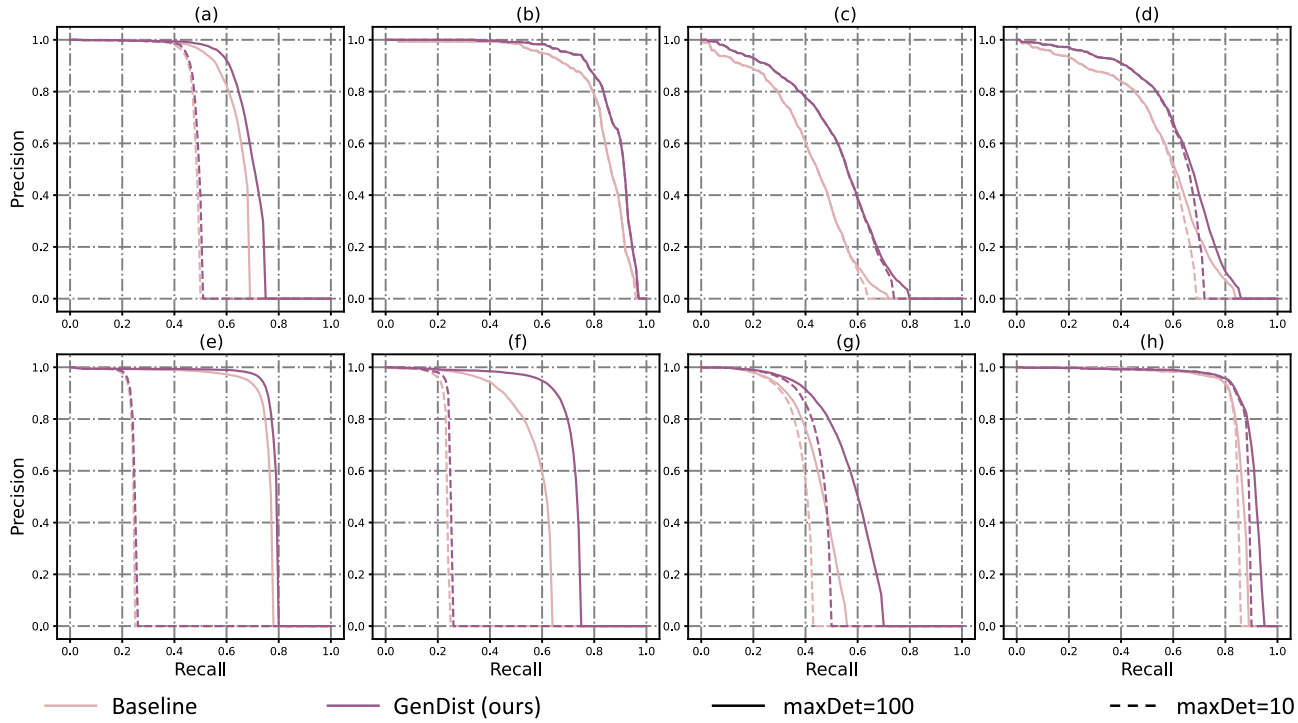


Fig. 9. Precision versus recall curves for GenDist and baseline for different IoU thresholds, categories, and maximum number of detections per image (maxDet). (a) PR curve @0.5, airplane. (b) PR curve @0.5, airport. (c) PR curve @0.5, bridge. (d) PR curve @0.5, harbor. (e) PR curve @0.5, ship. (f) PR curve @0.5, storagetank. (g) PR curve @0.5, vehicle. (h) PR curve @0.5, windmill.

Table X, the baseline model without a mask achieves 47.1 AP. For a comparison, we evaluate four different masking strategies: 1) random mask: masks generated through random sampling, with a 60% masking ratio; 2) grid mask: the 3×3 grid-based with randomly masked cells, maintaining the ratio at 60%; 3) center mask: masking only the central 30% of the ground truth box region; and 4) importance-based mask: guided by teacher features. The attention map from the teacher model is computed, values above the threshold (i.e., 1) are set to 0, and others to 1. All methods utilize L_2 loss as supervision, except the importance-based mask, which uses an additional L_1 loss. Finally, we evaluate the proposed O-SCM, which employs an adaptive strategy to determine the mask distribution. From the results, random and grid masks improve overall AP by 3.7 and 4.6 over the baseline, respectively, indicating the effectiveness of masking. The center mask, applied to the GT's central 30% region, attains 51.6 AP. In importance-based masking, the L_1 loss method lags 0.4 AP behind its homogeneous counterpart, suggesting L_2 loss is more suitable for KD. Ultimately, our method achieves optimal overall accuracy and small object detection performance, confirming that restoring the regions neglected by the student is more effective for KD.

5) Tradeoff Between Speed and Precision: We further evaluate the tradeoff between speed and accuracy across different methods. To quantitatively reflect the detectors' preference for different properties, we first calculate the normalized inference speed and accuracy, employing the teacher detector in Table IX as the reference point. Naturally, the larger the region defined by the coordinate point, the better the method balances speed and accuracy. The compared methods include the lightweight framework YOLOX [86], our proposed GenDist, and Faster R-CNN with Swin-base [79]. All of the models are trained using the multiscale strategy for 36 epochs, except for our proposed GenDist, which is trained for only 12 epochs. As shown in Fig. 8, although Faster R-CNN [79] achieves superior normalized performance than YOLOX [86], it usually comes at the cost of test speed. The expanded computational limits its applicability in smart devices. While YOLOX [86] surpasses our GenDist in inference efficiency, the gain is offset by a loss in precision. In contrast, our method achieves a favorable tradeoff between performance and efficiency, maintaining a high precision while offering competitive inference speed. Furthermore, the shorter training time contributes to greater deployment agility for our GenDist.

G. Model Analysis

1) Precision Versus Recall Curves: To obtain a clearer insight into GenDist's impact on detection performance, we visualize the precision–recall (P-R) curves for particular representative categories on DIOR [47] by the official toolkit. The P-R curve shape intuitively reflects the model's capability, better models typically yield larger areas under the curve (AUC) by preserving high precision at superior recall rates [39]. Fig. 9 illustrates that GenDist delivers stable performance improvements across different conditions, such as object categories, scales, and per-frame prediction. Remarkably, GenDist consistently enhances precision at all recall levels, particularly for the dense instances like "storagetank," "ship," and "vehicle," as well as the sparse ones such as "bridge" and

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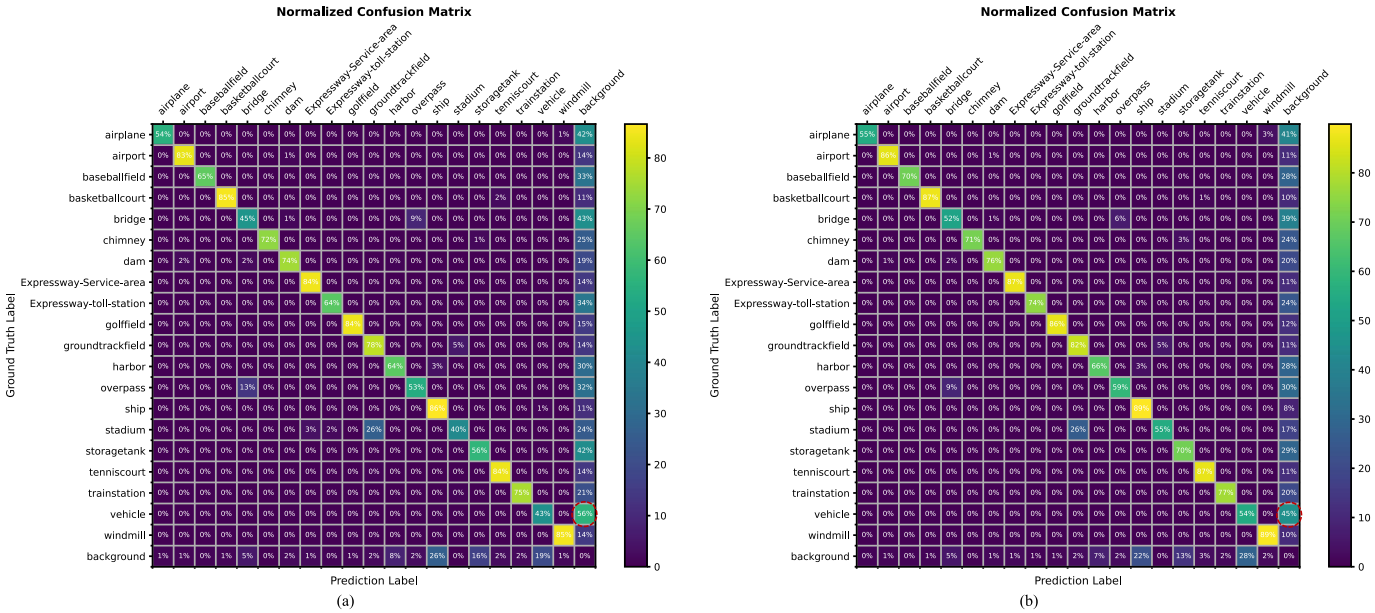


Fig. 10. Confusion matrix comparison of GenDist with baseline on DIOR. Class with markedly reduced missed detections is marked with red dashed lines. (a) Student. (b) GenDist (ours).



Fig. 11. Some typical failure predictions of GenDist on the DIOR dataset. The green and red bounding boxes indicate TP and FN predictions, respectively. Best viewed in color.

"windmill." In light of the preceding analysis, we contend that our GenDist has substantial potential for applications in disaster response and military contexts.

2) Confusion Matrix: We perform a systematic analysis by comparing the confusion matrices of GenDist and the student model. As shown in Fig. 10, the rows of the confusion matrix represent the ground truth labels, while the columns represent the predicted categories. Therefore, the last column indicates the miss rate for each category, and the bottom row indicates the false positive rate for each category. The diagonal values of the confusion matrix represent the prediction accuracy for each category. As presented in the results, the student achieves high accuracy for most categories but is prone to missing detections. For example, the miss rate for "bridge" that nearly blends with the background, along with compactly arranged "storagetank" and "vehicle," ranges from 40% to 60%. In contrast, GenDist reduces the miss rate by roughly 10% in these hard categories, making its predictions more reliable and comprehensive.

3) Limitation of GenDist: Despite the notable performance gains achieved by GenDist, object detection in optical remote sensing imagery continues to involve considerable challenges, particularly owing to the visual similarity between objects and the complexity of background settings. To objectively evaluate the limitations of GenDist, we present typical failure examples in this section. As illustrated in Fig. 11, our GenDist struggles to distinguish between "bridge" and "vehicle" that are closely connected to roads, due to blurred texture features. As depicted in Fig. 11, occluded small objects may lead to false negatives in our approach. In addition, certain "ships" located on land are not accurately detected. These findings mean that GenDist still has limitations in transferring contextual information.

4) Convergent Behavior: The convergence behavior of our method is shown in Fig. 12. From the results, the convergence process is stable. The generative loss $\mathcal{L}_{\text{dist}}$ (i.e., total loss minus detection loss) exhibits a rapid decrease in the first few epochs, facilitating the student to robustly reconstruct complete feature representations guided by teacher models. This masked feature recovery acts as a powerful regularizer, which in turn facilitates the convergence of the detection loss $\mathcal{L}_{\text{det}}$. Across the training

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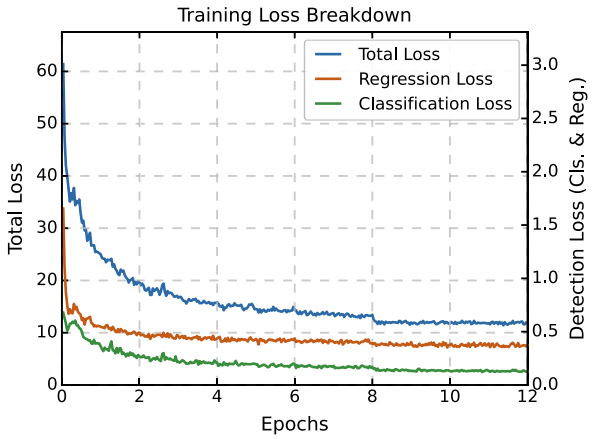


Fig. 12. Illustration of the convergence behavior of our GenDist. The training curves reflect the rapid early stage decline and stable optimization process of distillation and detection losses. Best viewed in color.

process, the two objectives operate cooperatively, without any observed optimization conflicts or gradient instabilities.

V. Conclusion

This work proposes a novel feature generation-based KD method, GenDist, refining the existing frameworks by three parts: mask generation, teacher feature refinement, and feature generation. First, we design an O-SCM that dynamically adjusts its distribution to guide the student model in reconstructing overlooked or misinterpreted feature regions. Second, we introduce DAGC to refine teacher features via context modeling and adaptive selection of core features, providing more accurate supervision signals for the student. Finally, through learnable offset sampling in SAG, we strengthen the reconstruction feature of small objects. Benefiting from the combined effect of all modules, our method achieves superior performance over state-of-the-art KD frameworks on the optical DIOR and SAR SSDD datasets. It also achieves competitive results on AI-TOD and SODA-D, both tailored for tiny object detection. Our GenDist maintains robust knowledge transfer even under challenging settings, e.g., limited training data and lightweight backbones. We expect GenDist to serve as a practical and efficient solution for remote sensing image processing.

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